MILITARY SPECIFICATION

RIOT CONTROL AGENT, CS2

1. SCOPE

1.1 This specification covers one type of riot control agent (hereinafter referred to as CS2).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Film (Polyethylene Thin Gage).
QQ-S-781	-	Steel Strapping, Flat.
RR-S-366	-	Sieve, Test.
TT-W-572	-	Wood-Preservative; Water-Repellent.
UU-S-48	-	Sacks, Shipping, Paper.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-D-729	-	Drums: Metal, 55-Gallon (For Shipment of Noncorrosive Material).
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bags, Interior Packaging.
MIL-B-131	-	Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-B-22085	-	Tape, Pressure-Sensitive Adhesive, Preservation and Sealing.
MIL-B-22205	-	Bags, Transparent, Flexible, Heat Sealable, for Packaging Applications.
MIL-C-51029	-	Chemical Agent, CS.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1233	-	Procedure for Determining Particle Size, Particle Size Distribution, and Packed Density of Powdered Materials.

DRAWINGS

US ARMY MUNITIONS COMMAND (EDGEWOOD ARSENAL)

D18-60-1283	-	Marking of Packing for Chemical Agent, CS-1 or CS-2.
E18-60-1294	-	Marking, Packing; Metal Shipping Container for Riot Control Agent CS, CS1 or CS2.
D18-60-1322	-	Palletization for Riot Control Agent (CS-1 or CS-2).
D150-2-2	-	Color Coding for Chemical Munitions.
C328-26-36	-	Bagged Agent, Assembly and Detail.

PUBLICATIONS

EDGEWOOD ARSENAL PURCHASE DESCRIPTION

196-21-87	-	Bottle, Screw-Cap, Plastic, 3-1/2 Gallon.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179	-	Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Raw materials. The raw materials used to prepare CS2 shall be as follows:

3.1.1 Chemical Agent, CS, ground. The CS used to prepare CS2 shall conform to the chemical and physical characteristics of MIL-C-51029. Prior to any blending operations, the CS shall be ground so that it will have an average particle size of no more than 7 microns when tested as specified in 4.4.4.3. (A maximum of 2 percent by weight of silica conforming to 3.1.2 may be added to the unground CS as a grinding aid provided that it is treated with 10 percent by weight of HMDS conforming to 3.1.3 during the final blending operation. In this event, instead of conforming to above average particle size requirement, the ground CS shall have an ultimate particle size no greater than 9 microns at 50-percent cumulative and no greater than 12 microns at 80-percent cumulative when tested as specified in 4.4.4.6.)

3.1.2 Silica (fumed colloidal silica). The silica used to prepare CS2 shall be Cabot "Cab-O-Sil HS-5."

3.1.3 Hexamethyldisilazane (HMDS). The HMDS used to prepare CS2 shall be Dow Corning "Z-6079 Silane," General Electric "SC-3726," Union Carbide "Y-7028," or Peninsular Chemresearch "02-34021-22."

3.2 Composition. CS2 shall be a uniformly blended mixture of 93.0 to 96.0 percent by weight of ground CS conforming to 3.1.1 and 4.0 to 7.0 percent by weight of HMDS-treated silica conforming to 3.2.1 when tested as specified in 4.4.4.1.

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3.2.1 HMDS-treated silica. The HMDS-treated silica used to prepare CS2 shall contain 80 to 90 percent by weight of silica conforming to 3.1.2 and 10 to 20 percent by weight of HMDS conforming to 3.1.3. No other ingredients shall be used to prepare the HMDS-treated silica. The HMDS-treated silica shall have an absorbance ratio of no less than 6.4 when tested as specified in 4.4.4.2.

3.3 Flow property. No less than 96.0 percent by weight of the CS2 shall pass through the test funnel in no more than 7.0 seconds when tested as specified in 4.4.4.4.

3.4 Apparent density. CS2 shall have an apparent density of no less than 0.20 gram (g) per cubic centimeter (cc) when tested as specified in 4.4.4.5.

3.5 Ultimate particle size. CS2 shall have an ultimate particle size no greater than 9 microns at 50-percent cumulative and no greater than 12 microns at 80-percent cumulative when tested as specified in 4.4.4.6.

3.6 Agglomeration. No more than 1.5 percent by weight of CS2 shall be retained on a No. 20 sieve and no less than 70 percent by weight of CS2 shall pass through a No. 325 sieve when tested as specified in 4.4.4.7.

3.7 Mechanical wetting resistance. CS2 shall retain its ability to re-aerosolize (form a dust cloud) when tested as specified in 4.4.4.8.

3.8 Hydrophobicity. CS2 shall have hydrophobic characteristics which yield a minimum slope of 0.33 milliliters (ml) per milligram (mg) when tested as specified in 4.4.4.9.

3.9 Preproduction. Prior to the start of regular production, a preproduction lot of CS2 shall be produced in accordance with this specification for test (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 500 pounds of CS2 shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production.

4.3.2 Sampling. Two representative 1-pound specimens of CS2 shall be randomly taken from the preproduction lot and placed in separate clean, dry, sealed containers labeled to identify the preproduction lot from which they were taken. The two specimens shall then be forwarded for tests to the test facility specified in the contract or order (see 6.1).

4.3.3 Inspection procedure for tests. Both preproduction specimens shall be separately tested as specified in 4.4.4, omitting 4.4.4.2 and 4.4.4.3. Failure of any test by either specimen shall be cause for rejection of the preproduction lot.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the CS2 offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.4.2 Sampling.

4.4.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling for test shall be conducted in accordance with table I. A representative specimen of no less than 1 pound shall be removed from each sample container and placed in a suitable clean, dry, closed container labeled to identify the lot and container from which it was taken (see 6.3).

Table I. Sampling for test

Number of containers in lot or batch	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.4.2.3 For raw materials.

(a) Ground CS. Two 50-gram (g) specimens shall be randomly taken from each ground batch of CS used to prepare CS2.

(b) HMDS-treated silica. Three 1-ounce specimens shall be randomly taken from each blended batch of HMDS-treated silica used to prepare CS2 and composited into one specimen.

4.4.3 Inspection procedure.

4.4.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined in accordance with MIL-STD-105 and the following classification of defects:

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Quantity per unit container incorrect
102	Container incorrect
103	Container damaged (punctured or torn)
104	Container leaking
105	Container closure incorrect
106	Outside of container contaminated with CS2
107	Marking missing, illegible, or incorrect
108	Palletization incorrect (where required)

4.4.3.2 For test. Each sample specimen taken in 4.4.2.2 shall be tested as specified in 4.4.4, omitting 4.4.4.2 and 4.4.4.3. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.4.3.3 For raw materials.

(a) Ground CS. Each sample specimen of ground CS taken in 4.4.2.3(a) shall be tested for particle size as specified in 4.4.4.3 or 4.4.4.6, as applicable. If either specimen fails the test, the batch of ground CS represented shall not be used to prepare CS2.

(b) HMDS-treated silica. The composited specimen of HMDS-treated silica taken in 4.4.2.3(b) shall be tested for absorbance ratio as specified in 4.4.4.2. If the composited specimen fails the test, the blended batch of HMDS-treated silica represented shall not be used to prepare CS2.

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4.4.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows (see 6.3):

4.4.4.1 Composition. Weigh approximately 10 g of specimen to the nearest 0.01 g into a tared crucible. Ignite the crucible contents to a constant weight at 700° C. Cool in a desiccator and reweigh. Calculate percent CS as follows:

$$\text{Percent CS} = 100 - \frac{100\ A}{W}$$

where: A = Weight of residue in grams and
W = Weight of specimen in grams.

Calculate the percent HMDS-treated silica as follows:

Percent HMDS-treated silica = 100 - percent CS.

4.4.4.2 Absorbance ratio of HMDS-treated silica. Using 4-inch by 4-inch by 0.5-millimeter (mm) silver chloride sheets (see 6.6), carefully punch out 13-mm discs if using the Perkin-Elmer macro die, or discs of the appropriate size to fit the plunger of the pellet die that is used. Make sure that any lubricant is first removed from the punch and die surfaces that will contact the silver chloride sheet. Conduct the operation under reduced light. Store the discs in the dark. Place the disc in the bottom of the pellet die. Add 7 to 8 mg of the HMDS-treated silica specimen. Tap the die on a table several times to obtain a uniform layer of the specimen on the disc. Without evacuation, press for 1 minute at 1000 pounds per square inch in a Carver Laboratory Press, Model B. Remove the pellet, and by means of an infrared spectrophotometer, scan quickly through the 3745 cm^{-1} and 2965 cm^{-1} regions to insure that the pellet is in the correct thickness (absorbance) range. Absorbance values between 0.5 and 1.0 are acceptable for quantitative work, with a desirable value of 0.6 to 0.7. If necessary, remake the pellet. Insert the pellet in a quartz spectrophotometer cell and condition the pellet in a vacuum at a temperature of 110° C for 1 hour. Cool to room temperature, align the quartz cell and pellet in the spectrophotometer, and obtain the spectrum between 4000 cm^{-1} and 1200 cm^{-1}. Determine the net absorbance, in absorbance units from the respective base lines, at 2965 cm^{-1} and 1860 cm^{-1}. Calculate the absorbance ratio as follows:

$$\text{Absorbance ratio} = \frac{\text{Net absorbance at 2965 } cm^{-1}}{\text{Net absorbance at 1860 } cm^{-1}}$$

4.4.4.3 Average particle size of ground CS. Determine the average particle size in accordance with MIL-STD-1233, method 100.

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4.4.4.4 Flow property. Transfer approximately 50 g of the specimen to a 1-pint mason jar. Pour the specimen from the jar to a second mason jar and repeat until the inner surfaces of both jars are coated with the specimen. Pass some of the specimen through a test funnel conforming to figure 1. Tap the jars and funnel to remove excess specimen. Tare weigh each jar with lid but not lid sealing ring. Place a primed receiver jar on a ring stand fitted with a 3-inch inside diameter ring, under the positioned ring. (A block of wood placed under the jar is suggested to facilitate subsequent removal.) Tentatively rest funnel in ring. Into the second jar, weigh 20 + 0.1 g of specimen. With one hand, hold the jar and lid in place and shake vigorously for approximately 10 seconds to fluidize the specimen, remove the lid, cover the jar with the test funnel, and immediately invert the jar-funnel assembly over the receiver jar, allowing the funnel to rest on the ring and remove the empty specimen jar. (The time lapse between the shaking of the capped jar and the inversion of the jar-funnel assembly should be kept to a minimum to prevent settling of the fluidized specimen. The flow of fluidized specimen through the funnel must be by gravity alone; shaking the assembly in any way to assist flow is not permitted.) Time the pouring process. Start the time as soon as the first material touches the funnel. Stop the time when all material has passed the funnel discharge or when the material stops flowing, whichever occurs first. Record the time in seconds. Remove the receiving jar, replace the lid, and weigh. Tap the residue on the funnel into another tared jar and weigh. Calculate the percent passing through test funnel as follows:

$$\text{Percent passing through test funnel} = \frac{100\ A}{(A + B)}$$

where: A = Weight of specimen passing through funnel in grams and
B = Weight of specimen tapped from test funnel in grams.

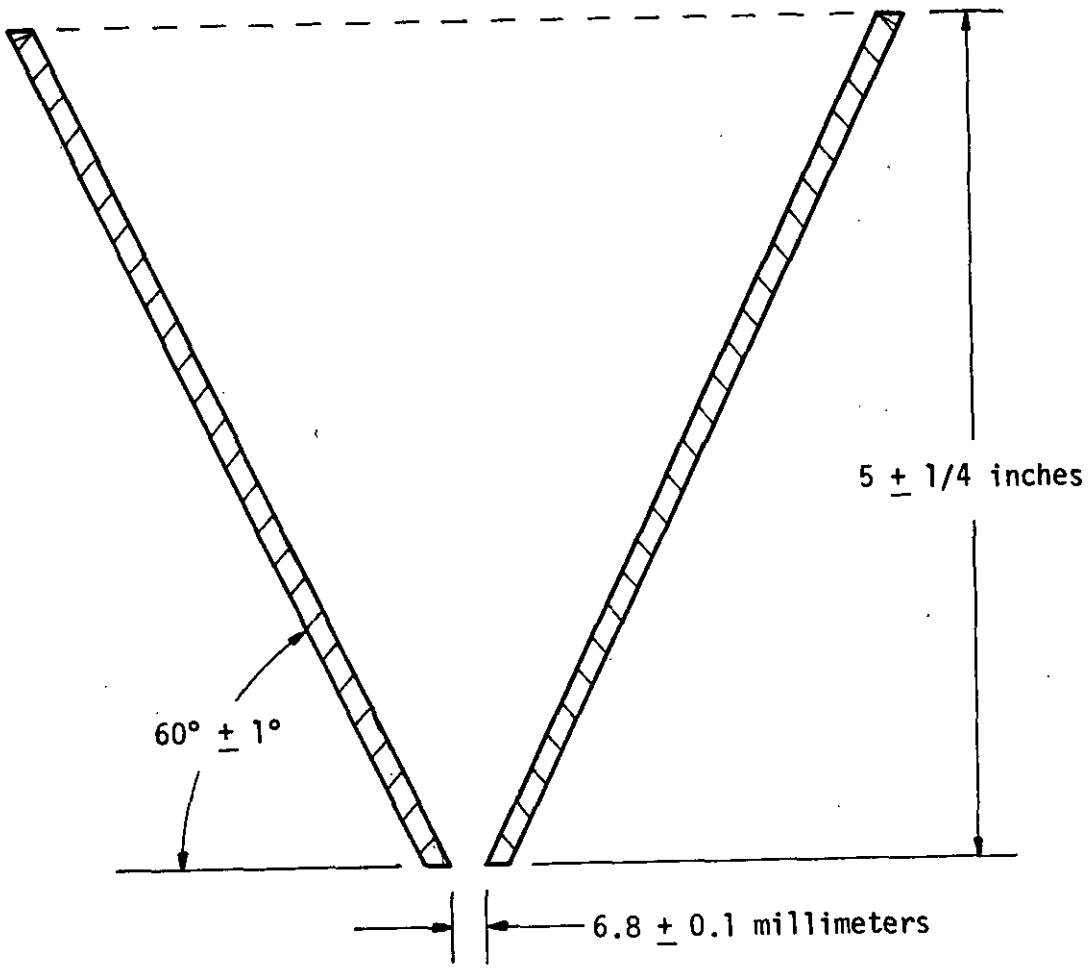
The total of A + B shall be no less than 19 g for the test to be valid.

4.4.4.5 Apparent density. Add 20 + 1 g of specimen, by gravity feed, to a tared 100- or 150-ml graduated cylinder. Let the specimen settle undisturbed for 10 minutes, read the fill volume to the nearest milliliter graduation, and record. Weigh the cylinder and specimen. Calculate apparent density in grams per cubic centimeter as follows:

$$\text{Apparent density (grams per cubic centimeter)} = \frac{A - B}{V}$$

where: B = Tare weight of cylinder in grams,
A = Weight of cylinder and specimen in grams, and
V = Volume of specimen in milliliters.

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Material: Rigid vinyl acetate, planished, 1/8-inch thick.

Construction: Vacuum formed, apex filed or ground to specified orifice. Orifice through apex shall not be drilled.

Figure 1. Test funnel for flow property

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4.4.4.6 Ultimate particle size.

4.4.4.6.1 Apparatus.

(a) Microscope equipped as follows:

(1) Compensating eyepiece, 10X.

(2) Apochromatic objective, 43X (4 mm) with a 0.95 numerical aperture, and a correction collar for cover glass thickness variation.

(3) Aplanatic apochromatic sub-stage condenser of high numerical aperture.

(4) Stage micrometer, consisting of a reticle mounted on a standard 1- by 3-inch glass slide. The reticle has a millimeter scale divided into 100 divisions, each scale division being equivalent to 10 microns.

(5) Microscope slides, glass, standard 1- by 3-inch.

(6) Cover glasses, No. 1, 3/4-inch diameter.

(7) Projection equipment and screen. A right-angled eye-piece lens may be used together with a projection screen or a ground glass screen. The distance between the projection screen and the microscope must be capable of being varied.

(b) Transparent ruler, graduated in millimeters.

(c) Hand magnifying glass for better viewing when measuring the particles projected on the screen.

(d) Eye-dropper.

(e) Logarithmic probability graph paper.

(f) Beaker, 200-ml capacity, tall form.

(g) Light mineral oil.

(h) Stirrer, electric, operating at 1650 ± 150 revolutions per minute, and having twin blades not exceeding a total of 1.7 inches in diameter.

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4.4.4.6.2 Calibration. Place the micrometer slide on the stage of the microscope in the normal position. Project the image on the screen. Adjust the distance between the microscope and the screen until 67 millimeter divisions of the transparent ruler are equal to 10 divisions of the image on the screen. One division of the ruler is now equal to 1.5 microns.

4.4.4.6.3 Procedure.

(a) Preparation of sample on slides. Weigh approximately 0.100 g of specimen into each of 2 beakers. To one, add 20 ml of light mineral oil and mix with the electric stirrer for 5 minutes ± 30 seconds. While stirring, withdraw a portion of the mixture with an eye-dropper from a point approximately one-fourth the diameter of the beaker, measured from the inside wall of the beaker and just above the top of the stirrer blades. Immediately wipe off the outside of the eye-dropper and deposit one drop of the mixture on a clean, dry microscope slide. Cover the drop with a cover glass. Prepare a second slide in the same manner. Follow the same procedure with the second beaker and prepare a second pair of slides.

(b) Measurements. Mount one of the slides on the microscope stage and project the image. Place the transparent ruler horizontally across the image so that the field is approximately bisected. Determine, to the nearest millimeter, the maximum length of the horizontal projection of each particle successively encountered regardless of the orientation of the particle, excluding clusters of particles or agglomerates. When the edge of the field of vision is reached, continue measurements parallel to the above measurement but sufficiently distant so that none of the previously counted particles are measured. Continue traversing the field completely and measuring particles until at least 200 particles having a length of over 1.5 microns are counted. A hand magnifying glass may be used to assist in the measurement of the particles. If one-fourth or more of the field of vision is occupied by clusters of particles (agglomerates), discard the sample in that beaker and prepare a replacement for determination. Disregard any results from that beaker and use the results from a replacement beaker in which the particles are not agglomerated when averaging the results from the four slides. Photomicrograph techniques are permitted in the particle counting in lieu of screen projection.

(c) Tabulation of frequency distribution. As the particles are measured, tally them into class intervals as follows:

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Class interval in millimeters	Midpoint of class in microns	Frequency (assumed as an example)
0.0- 0.5	---	Disregard
0.6- 1.5	1.5	87
1.6- 2.5	3.0	115
2.6- 3.5	4.5	59
3.6- 4.5	6.0	26
4.6- 5.5	7.5	14
5.6- 6.5	9.0	6
6.6- 7.5	10.5	1
7.6- 8.5	12.0	1
8.6- 9.5	13.5	0
9.6-10.5	15.0	0
10.6-11.5	16.5	0
11.6-12.5	18.0	0

(d) Calculation of particle size. After recording and totaling the number of particles in each micron class, calculate the percent cumulative as follows (see table II):

$$\text{Percent cumulative} = \frac{nd^3 \text{ cumulative} \times 100}{\text{total } nd^3}$$

On logarithmic probability graph paper plot the class size in microns on the logarithmic scale and the percent cumulative by weight on the other scale. From the curve, determine the particle size at the 50-percent cumulative intercept and at the 80-percent cumulative intercept. Repeat the procedure for each of the remaining slides. Average the results obtained. Table II shows the recording of typical data.

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Table II. Sample data

d	n	d^3	nd^3	nd^3 cumulative	Percent cumulative by weight
1.5	87	3.375	293	293	1.0
3.0	115	27.00	3,100	3,393	12.2
4.5	59	91.13	5,380	8,773	31.6
6.0	26	216.0	5,620	14,393	51.8
7.5	14	421.9	5,900	20,293	73.0
9.0	6	729.0	4,360	24,553	88.4
10.0	1	1,000.0	1,000	25,553	92.1
12.0	1	2,197	2,197	27,750	100.0
13.5	-	2,460	-	-	-
15.0	-	3,375	-	-	-
16.5	-	4,492	-	-	-
18.0	-	5,832	-	-	-

d = midpoint of class in microns.
n = number of particles tallied in micron class.

A plot of the sample data shows that the particle size equals 5.5 microns at 50-percent cumulative and 8.3 microns at 80-percent cumulative.

4.4.4.7 Agglomeration. Use sieves conforming to RR-S-366. Nest a No. 20 sieve on top of a No. 325 sieve and place a receiving pan on the bottom and a cover on top. Weigh to the nearest 0.1 g approximately 50 g of the specimen and quantitatively transfer to the No. 20 sieve. Tape the outside of the assembly and place it in a mechanical shaker geared to produce 300 ± 5 gyrations and 150 ± 10 taps of the striker per minute. Vibrate for 20 minutes with the tapper in operation. Weigh the specimen retained on the No. 20 sieve and calculate the percent retained. Weigh the specimen on the bottom pan and calculate the percent passing through the No. 325 sieve.

4.4.4.8 Mechanical wetting resistance.

(a) Apparatus. Use a wrist action shaker adjusted to a vibration amplitude of 3/4 ± 1/16 inch and a vibration frequency of 300 ± 10 oscillations per minute at a distance of 2-1/4 ± 1/8 inch from the center of oscillation (see 6.5).

(b) Procedure. Weigh 0.50 g of the specimen into a 2-inch diameter, 4-ounce wide mouth jar. Add 70 ml of water having a temperature of 75° ± 5° F. Cap the jar and position it in a shaker having the characteristics specified in (a). Shake the jar for 40 ± 0.5 minutes.

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Remove the jar from the shaker and shake manually several times. Remove the cap and using a contrasting background, observe any re-aerosolization (dust cloud formation) of the specimen.

4.4.4.9 Hydrophobicity.

(a) Standard detergent solution. Prepare a solution of "Sparkleen" detergent (see 6.4) in water at a concentration of 10 g per liter. Allow to stand for 24 hours at room temperature. Transfer 200 ml of the solution to a 1-liter volumetric flask and dilute to the mark with water. This solution contains 2 g of "Sparkleen" per liter.

(b) Procedure. Weigh to the nearest 0.1 mg approximately 0.015 g of the specimen into a 100-ml graduated cylinder fitted with a ground glass stopper. Carefully add 20 ml of water, stopper, and shake vigorously. (At this point the specimen should be "crawling" up the walls of the cylinder. Some particles may appear to be wetted but the majority of the specimen should remain at the surface of the water.) Add from one-fourth to one-third of the weight of the specimen in milliliters of the standard detergent solution prepared as specified in (a). (For example, if the specimen weighs 36.0 mg, 9 to 12 ml of detergent solution would be added.) Agitate by inverting the cylinder. When adding detergent, do not shake. Invert the cylinder two or three times, making sure that the detergent solution is evenly distributed throughout the mixture and that it comes in contact with all of the specimen. (At this point, the specimen should no longer "crawl" up the walls of the cylinder. The solution should be cloudy and after 1 minute, one-third to one-half of the specimen should be at the top of the liquid. Any suds that had formed initially upon inversion should rapidly disappear.) Add more of the standard detergent solution in 4-ml aliquots until the suds formed fail to degenerate after standing for 2 mintues. Then add the standard detergent solution in 2-ml aliquots until the suds remain stable for 5 minutes and there is a definite tendency for the particles to settle. (At the end of 5 minutes of standing, about 2 to 6 ml of solution directly beneath the surface may be relatively clear of particles.) Verify the end point by adding an additional 2-ml aliquot of standard detergent solution and then agitating. If at the end of 5 minutes, the characteristics of the solution have not changed, the end point is verified. If the end point has been verified, add an additional 6-ml aliquot of the standard detergent solution and agitate. If at the end of 5 minutes, the characteristics of the solution have not changed, the end point has been reached. Repeat the entire procedure using 0.030 g and 0.045 g of the specimen weighed to the nearest 0.1 mg.

(c) Calculations. Plot on coordinate paper, milligrams of specimen on the x-axis versus milliliters of standard detergent solution added on the y-axis. Determine the slope of the line formed by the plotted points as follows:

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$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

where: x_2 and x_1 = Weights of specimen in milligrams and
y_2 and y_1 = Milliliters, respectively, of standard detergent solution added.

NOTES

1. When the 20 ml of water is added to the specimen, the specimen is shaken as a check. If the specimen is immediately wetted or mostly wetted (70 to 90 percent), the cylinder may not have been clean and the test should be repeated using a clean cylinder. If the same behavior is again observed, it is probable that the specimen is very poor and the standard detergent solution should be added in 2-ml aliquots from the beginning.

2. The time required from the addition of the first aliquot of standard detergent solution to the end point usually is 10 to 20 minutes. The maximum time allowed is 30 minutes since the specimen will appear to be less hydrophobic than it is if it is allowed to remain in the presence of the standard detergent solution for a longer period of time. If the specimen is tested in less than 10 minutes, it will appear to be more hydrophobic than it is.

3. After a test is completed, 5 to 10 ml of acetone should be added to the specimen solution and the mixture should be shaken. The acetone will aid detoxification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Packaging shall be level A for CS2 in 8-pound bags, 8-pound plastic bottles, or 1/3-pound bags as specified (see 6.1).

5.1.1 Eight-pound bags. Eight pounds + 2 ounces of CS2 shall be packaged in either a 17-1/2-inch by 25-inch bag or a 16-inch by 24-inch bag conforming to type I, class E, style 1 of MIL-B-117. The bags shall consist of four plies with the outer ply of scrim-back being bonded to polyethylene-aluminum foil-polyethylene laminate. Each bag shall be closed by heat sealing and shall then be inserted into a snug-fitting sack conforming to type II, construction No. 1 of UU-S-48. The sack shall be closed by stitching as specified in UU-S-48 without stitching or otherwise damaging the inner heat-sealed bag. The outer surfaces of the bag and the sack shall be free of CS2 contamination.

5.1.2 Eight-pound plastic bottles. Eight pounds + 2 ounces of CS2 shall be packaged in a plastic bottle conforming to Edgewood Arsenal Purchase Description 196-21-87 and in accordance with any additional requirements specified in the contract or order (see 6.1). The tightened

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screw cap closure shall be further secured to the neck of the bottle with tape of 2-inch minimum width conforming to type II of MIL-T-22085. The tape shall be tightly wound around the neck of the bottle and the cap for a minimum of two complete turns. The tape shall extend from the top of the cap to a circumferential line around the bottle spout no less than 3/4 inch below the bottom of the tightened cap. The plastic bottle and closure shall be free of CS2 contamination.

5.1.3 One-third-pound bags. CS2 shall be packaged in a 0.35 ± 0.035 pound quantity in a kraft paper bag as shown on Drawing C328-26-36.

5.1.4 Intermediate packaging for 1/3-pound bags. Twenty-five bags of CS2, packaged as specified in 5.1.3, shall be intermediately packaged in a bag conforming to type I, class E of MIL-B-117 which shall then be closed by heat sealing. Included air volume shall be held to a practical minimum.

5.2 Packing. Packing shall be level A or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Eight-pound bags. Ten bags of CS2, packaged in sacks as specified in 5.1.1, shall be packed in a 55-gallon drum having a fully removable head. The drum shall have an inside diameter of 22-1/2 ± 1/16 inch, measured 1/2 inch down from the top of the drum, and a minimum inside height of 34-3/8 inches measured on the vertical axis of the drum between the bottom head and a diameter across the top of the upper curl. The drum shall conform to Department of Transportation (DOT) Specification 5B or 6C and shall be cleaned and painted as specified in PPP-D-729. The interior arrangement shall consist of two tiers of five sacks per tier. Tiers shall be separated by fiberboard pads. Additional blocking in the form of fiberboard cells shall be used as required to assure a tight pack. Fiberboard shall conform to grade W5c of PPP-F-320.

5.2.1.2 Eight-pound plastic bottles. Five plastic bottles of CS2, packaged as specified in 5.1.2, shall be packed side by side, in an upright position, with pouring spouts facing in alternate directions in a close fitting fiberboard box. The fiberboard box shall be enclosed in a heat-sealed bag and placed in a nailed wood box in accordance with submethod IA-14 of MIL-P-116. The fiberboard box shall conform to style RSC, type CF, class domestic, variety SW, grade 125 of PPP-B-636. Fiberboard separators conforming to type CF, class domestic, variety SW, grade optional of PPP-F-320 shall completely separate the plastic bottles and assure a tight pack. A flat fiberboard pad, extending the full length of the box shall be placed above the handles and between the spouts. The pad shall extend 1/4 ± 1/8 inch above the closure of the spout. The fiberboard for the pad shall be of the same material as that specified for the separators. The fiberboard box shall be closed

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and sealed in accordance with the requirements in the appendix to PPP-B-636. The heat-sealed bag enclosing the fiberboard box shall be formed from barrier material conforming to class 1 of MIL-B-131. The nailed wood box enclosing the fiberboard box and heat sealed bag shall conform to class 2, style 4-1/2, for a type 3 load exceeding 100 pounds but not exceeding 250 pounds of PPP-B-621. Fiberboard pads conforming to grade W5c or W6c of PPP-F-320 shall be inserted between the heat-sealed bag and all six interior surfaces of the wood box. The wood box shall be closed and strapped with galvanized strapping conforming to type I, class B of QQ-S-781. The box or the finished wood parts thereof shall be completely immersed for no less than 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.1.3 One-third-pound bags. Ten intermediate packages of CS2, packaged as specified in 5.1.4, shall be packed in a 55-gallon drum as specified for the eight-pound bags in 5.2.1.1.

5.2.2 Level C.

5.2.2.1 One-third-pound bags. Approximately 250 1/3-pound bags of CS2, packaged as specified in 5.1.3, shall be packed in a steel drum conforming to DOT Specification 5B or 6C in accordance with DOT regulations. Each drum shall be lined with a heat-sealed bag formed from polyethylene of 0.004-inch minimum thickness conforming to type I of L-P-378.

5.2.2.2 Bulk quantities. A quantity of 125 ± 1 pound of CS2 shall be packed in a 55-gallon fully-removable head drum conforming to DOT Specification 5B. The drum shall be lined with a heat-sealed bag conforming to type III of MIL-B-22205. Care shall be taken so that no CS2 is deposited between the drum and the liner.

5.3 Marking.

5.3.1 Eight-pound bags. Each eight-pound bag shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked as shown on Drawing E18-60-1294.

5.3.2 Eight-pound plastic bottles. Each eight-pound plastic bottle shall be marked in accordance with MIL-STD-129. In addition the centers of the diagonally opposite vertical edges to the right of the identification marking shall be color coded with two red stripes on a gray background and marked with the identification symbol "CS2" in accordance with Drawing D150-2-2. Shipping containers shall be marked in accordance with Drawing D18-60-1283.

5.3.3 One-third-pound bags. No marking is required for 1/3-pound bags. Intermediate packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing E18-60-1294.

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5.4 Palletization. When specified in the contract or order (see 6.1), shipping containers of CS2, packed as specified in 5.2.1.2, shall be palletized in accordance with Drawing D18-60-1322.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1).
(c) Additional requirements for 8-pound plastic bottles, if required (see 5.1.2).
(d) Level of packing required for 1/3-pound unit quantities (see 5.2).
(e) Palletization, if required (see 5.4).
(f) Preproduction.

(1) Time allowed for supplier submission of sample specimens for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.3 Caution. Tests should be performed in a hood. Attention is called to Chemical Corps Safety Directive SD 385-12.

6.4 "Sparkleen" detergent. "Sparkleen" detergent is available from Fisher Scientific Company.

6.5 Wrist action shaker. A Burrell, Model BT shaker, available from the Burrell Corporation, Pittsburgh, Pennsylvania, has been found satisfactory for the mechanical wetting resistance test.

6.6 Silver chloride sheets. Silver chloride sheets for the absorbance ratio of HMDS-treated silica test may be obtained from the Harshaw Chemical Company, 1945 E. 97th Street, Cleveland, Ohio 44106.

6.7 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

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6.8 Supersession data. This specificaton includes the requirements of Edgewood Arsenal Purchase Description 196-21-84A dated 19 April 1968, and Amendment 1 dated 6 June 1968.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1365-A045

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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